

SPOKE Visualizations

Envisioning Spoke & User Studies

Indiana University Team

- Katy Börner: CNS Director & MC-IU PI
- Andreas Bueckle: Research Scientist and Postdoc
- Bruce Herr II: Senior Systems Architect and Project Manager
- Elizabeth Maier: User Experience Designer



Envisioning SPOKE: 3M Nodes and 30M Edges

The Scalable Precision Medicine Oriented Knowledge Engine (SPOKE) graph federates about 19 open datasets into a public data commons of health relevant knowledge. This site lets users explore the massive SPOKE knowledge graph.

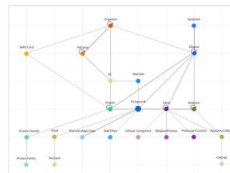
The site was designed for two user groups: (1) novice users interested to understand the coverage and quality of SPOKE data and (2) expert users interested to analyze and optimize the interlinked knowledge graphs in SPOKE.

The overview visualization shows the different entity type and their diverse interlinkages. Detail visualizations show

This project is funded by NSF award 2033569.

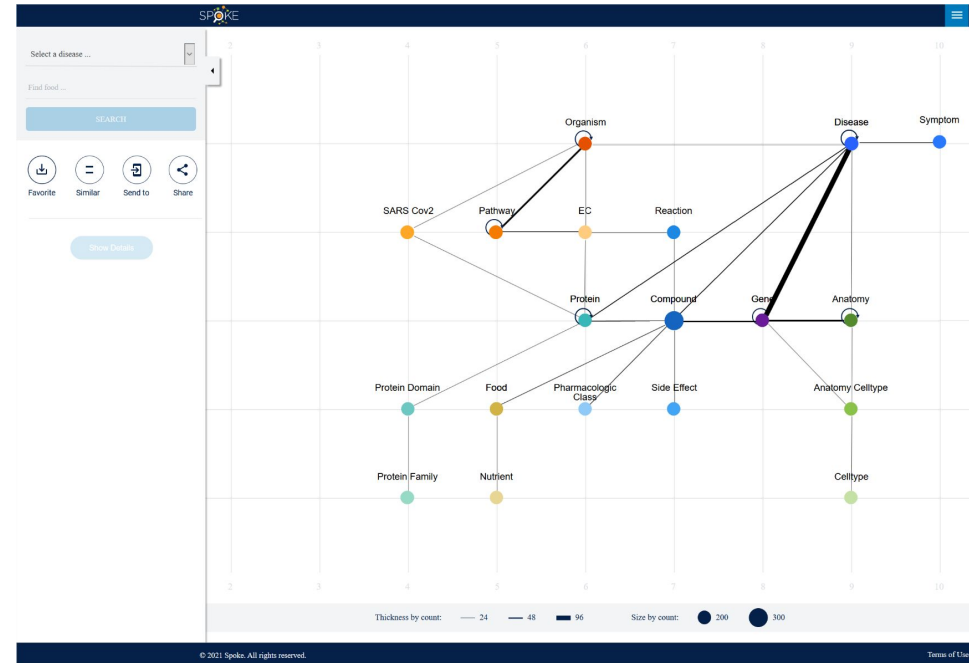
SPOKE is a fully interactive tool for exploring the interconnections between data.

Explore SPOKE



Envisioning SPOKE

- Federates many open datasets into a public data repository of health-relevant knowledge
- Lets users explore the massive SPOKE knowledge graph.
- Designed for two user groups:
 - Novice** users interested to understand the coverage and quality of SPOKE data
 - Expert** users interested to analyze and optimize the interlinked knowledge graphs in SPOKE.
- The overview visualization shows the different entity type and their diverse interlinkages.
- Fully interactive tool for exploring the interconnections between data.





<https://cns-iu.github.io/spoke-vis>

Select disease ...

coronary artery disease

Search for food ...

Atlantic salmon

Search

Show Details

This metagraph view will
be used in the
SPOKE-BRIC application.



Select disease ...

coronary artery disease

Search for food ...

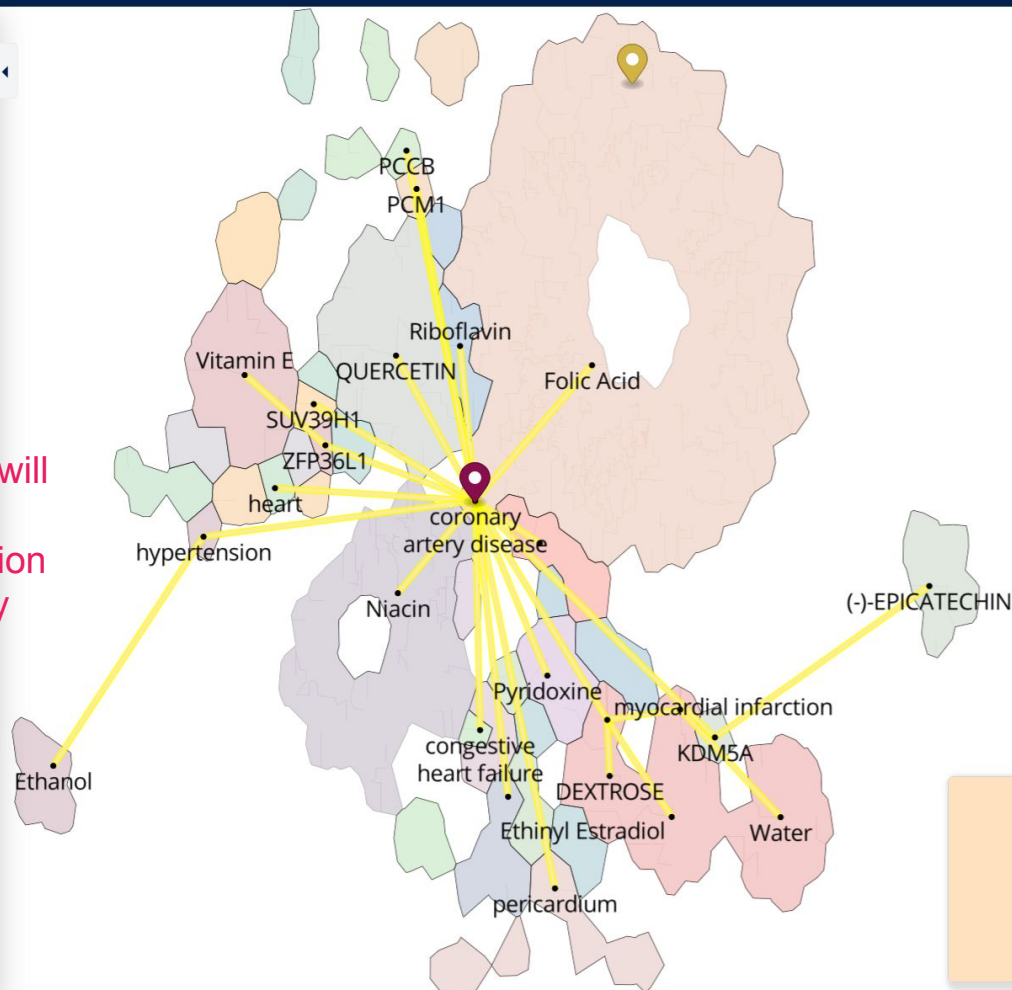
Atlantic salmon

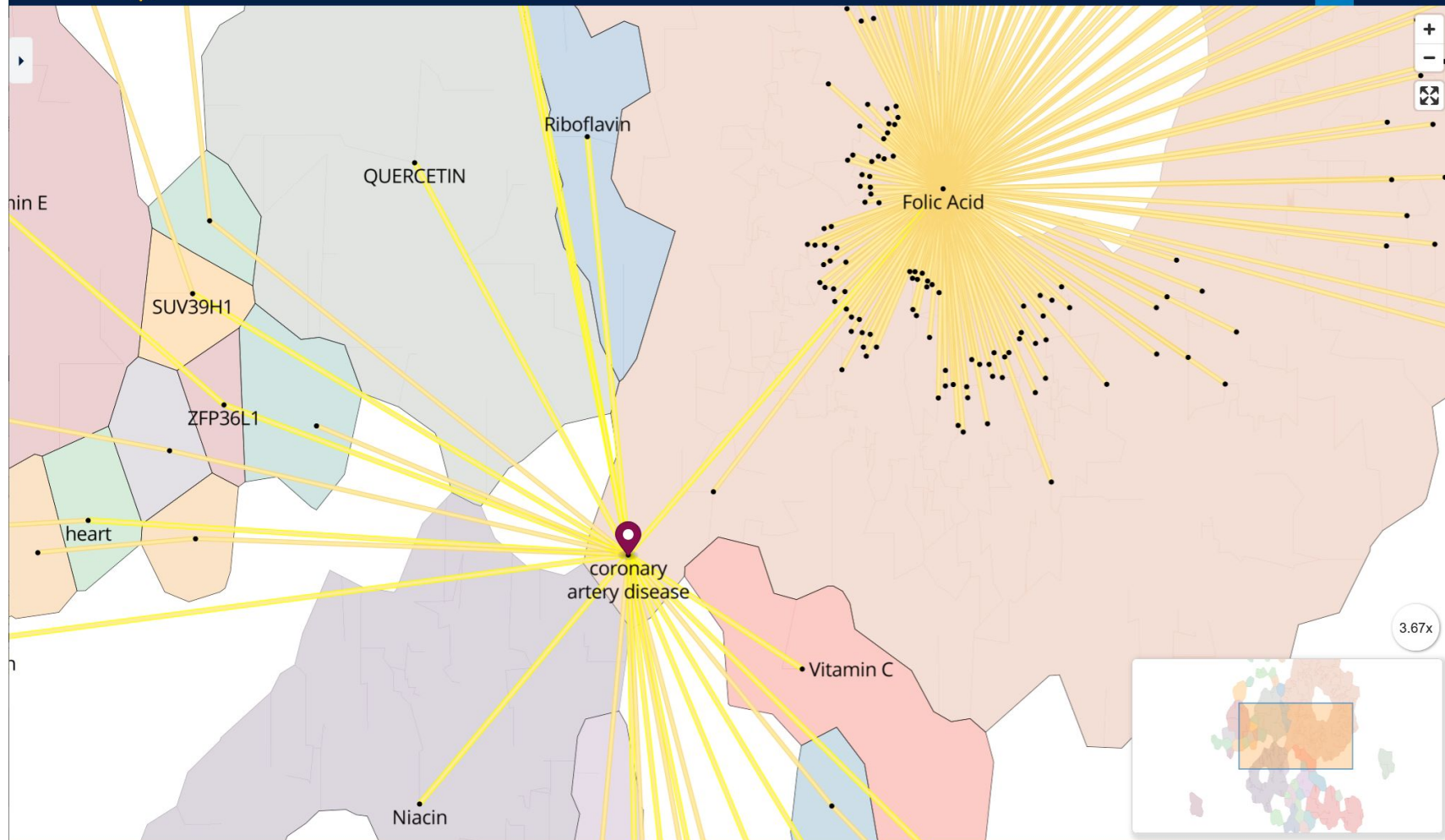
Search

Show Overview

This kind of map view will be used in the SPOKE-BRIC application (renamed into ontology explorer).

- 897 foods
- 10,209 diseases





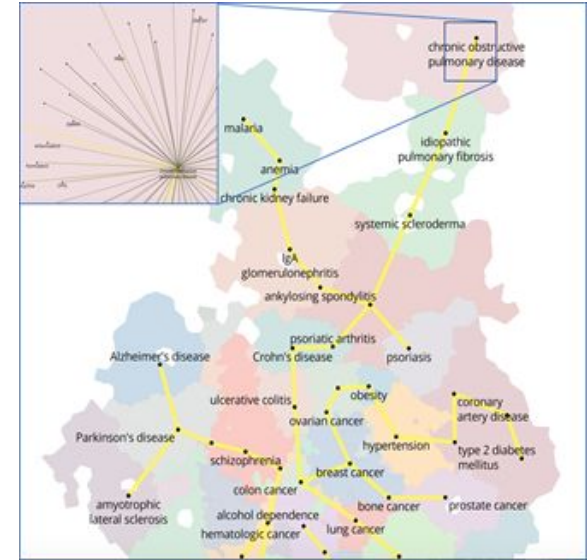
Envisioning SPOKE

SPOKE UI implements interactive, multi-level visualizations of large networks.

Standard network visualizations of large real-world networks often resemble “hairballs” that provide little actionable insight.

SPOKE visualizations compute and display clusters of related nodes and backbones between major nodes at each level of detail. The visualizations resemble geospatial maps at mid-fidelity resolutions with continents of similar nodes and real paths (backbones) for each level, similar to geographic maps that show real cities and real roads at every level of detail.

Previous user studies comparing network vs. map-like visualizations revealed that map-like visualizations are easier to read, remember, and communicate.



User Study: Design

- Two user cohorts
 - Patients with undiagnosed diseases
 - Non-patients
- Exclusion criteria: age 18+
- IU IRB #2011596050
- Study workflow
 - Pre-Q with Qualtrics
 - Task phase on SPOKE website
 - Post-Q with Qualtrics
 - User interaction captured with Google Analytics
 - Data queried and pulled for analysis Using Big Query
 - responseld and cohort tag from Qualtrics to track users between Qualtrics and Google Analytics

Processing location: US

Query results [SAVE RESULTS](#) [EXPLORE DATA](#)

Query complete (1.2 sec elapsed, 1.4 MB processed)

Job information [Results](#) JSON Execution details

8	https://611be6cab3acde087ccca3ba-spoke-vis.netlify.app/	
9	http://localhost/	
0	https://cns-iu.github.io/spoke-vis/visualization/details?disease=cad&food=Beer	
1	https://cns-iu.github.io/spoke-vis/visualization/details?disease=epilepsy&food=Salmonidae%20(Salmon,%20Trout)	
2	https://cns-iu.github.io/spoke-vis/visualization/overview?disease=epilepsy&food=Salmonidae%20(Salmon,%20Trout)	
3	https://cns-iu.github.io/spoke-vis/visualization/details?disease=epilepsy&food=Atlantic%20salmon	
4	https://cns-iu.github.io/spoke-vis/visualization/details?disease=cad&food=Abalone	
5	https://cns-iu.github.io/spoke-vis/visualization/details?disease=cad&food=Condensed%20milk	
6	https://cns-iu.github.io/spoke-vis/visualization/details?disease=epilepsy&food=Chili	
7	https://cns-iu.github.io/spoke-vis/visualization/overview?disease=alzheimers-all&food=Abiyuch	
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3	https://cns-iu.github.io/spoke-vis/visualization/details?disease=cad&food=Acerola	
4	https://cns-iu.github.io/spoke-vis/home?responseld=Q12345678&cohort=non_patient	
5	https://cns-iu.github.io/spoke-vis/visualization/details?disease=cad&food=Rainbow%20smelt	
6	https://cns-iu.github.io/spoke-vis/home?queryString=IAmAUser	
7	https://cns-iu.github.io/spoke-vis/visualization/details	
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9	https://cns-iu.github.io/spoke-vis/visualization/overview?disease=cad&food=Butter	
0	https://cns-iu.github.io/spoke-vis/visualization/overview?disease=cad&food=Potato%20gratin	
1	https://cns-iu.github.io/spoke-vis/visualization/overview?disease=cad&food=Coffee	

Research Questions

RQ1: Is there a difference in engagement between patients and non-patients, measured by:

- time spent,
- number of clicks, (events when edge of node is clicked)
- amount of mouse movement, (mousemove event)
- number of search terms, (search_click)
- number of different visualization states (overview vs detail: show_overview_clicked, show_details_clicked)
- amount of interaction techniques used (e.g., zoom_change + zoom value)
- Focus of camera (=> pan_change)
- Number of edges clicked (edge_clicked)
- Time spent in overview vis vs. explorer? (timestamp, event_name)
- Number of different search term combinations in overview vis? (page_location)

H1: Patients have measurably more engagement with the visualization than non-patients (defined via metrics from interaction data, e.g., more zooming, more dropdown menu changes, more clicks, more mouse movement). Or maybe low scores and no diff => bad for vis; or high scores and no diff => good for vis!

RQ2: Is there a difference in task score between the two groups? Example tasks:

- Which compound connects Atlantic Salmon with epilepsy?
- Is beer advantageous or disadvantageous towards Alzheimer's disease?
- Can you find five foods that treat coronary artery disease?

H2: There is no significant difference in task score, meaning that pre-existing knowledge has no bearing on the ability to answer questions with the visualization. Prior expertise in data visualization also makes no difference.

Power Analysis

We are using G*Power¹ to understand the number of users to run to get a certain statistical power.

Two user cohorts

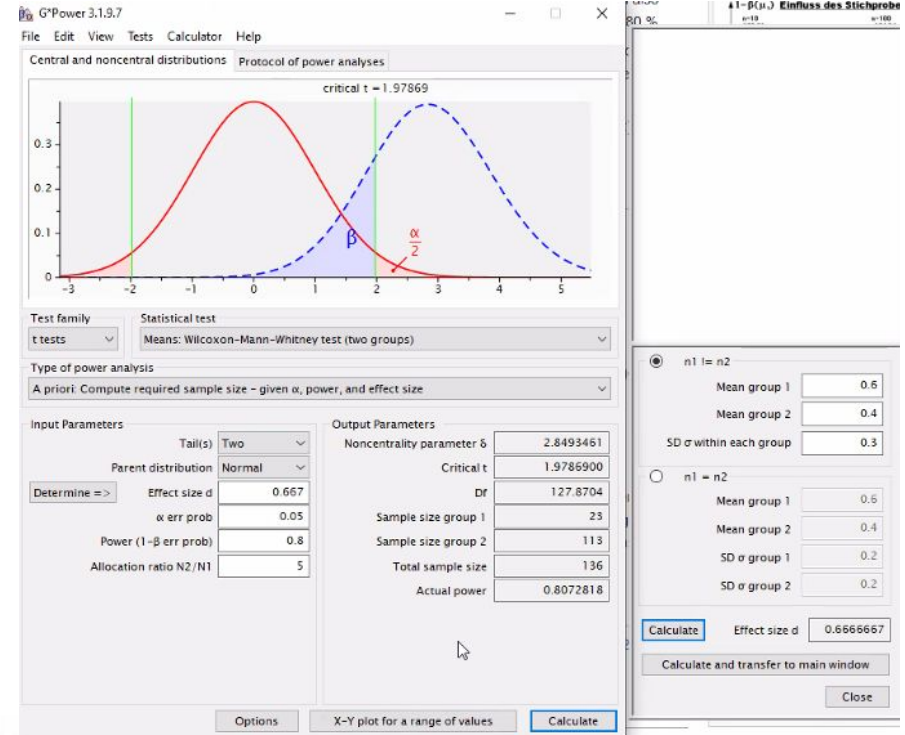
- Patients with undiagnosed diseases
- Non-patients

Exclusion criteria: age 18+

IU IRB #2011596050

Parameters

- Two-tailed t-test
- Effect size: 0.667
- Power: 0.8 (standard value)
- Ratio between cohorts: 5:1
- Need **25 patients**
- Need **113 non-patients**



¹ F. Faul, E. Erdfelder, A.-G. Lang, and A. Buchner, "G* Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences," Behavior Research Methods, vol. 39, no. 2, pp. 175-191, 2007, doi: 10.3758/bf03193146.

Survey

- Built using Qualtrics
- We set a cohort tag via the survey URL (query string)
- Pre-Q about vis expertise
- Demographics
- Tutorial + exploration prompt
- Tasks
- Post-Q to capture feedback
- Pilot study: 8 subjects

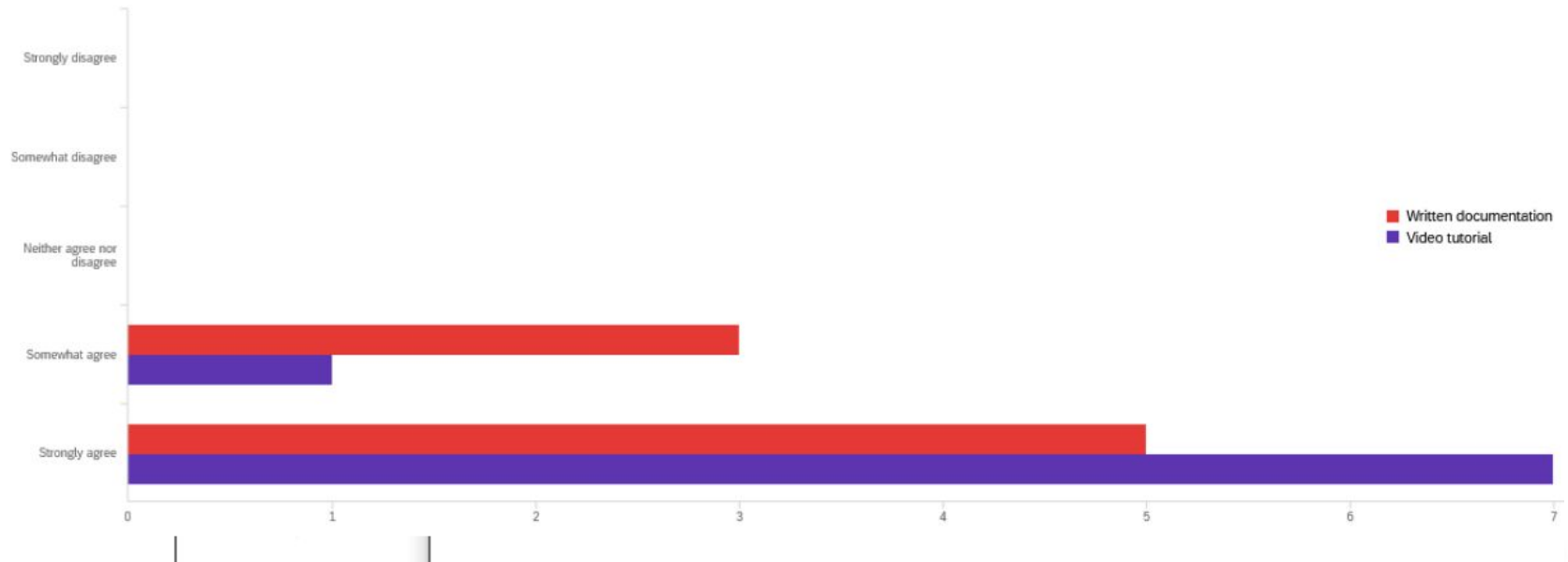
The screenshot shows the Qualtrics survey editor interface. At the top, there is a 'Set Embedded Data:' section with a 'cohort' tag and a note 'Value will be set from Panel or URL... Set a Value Now'. Below this, there is a list of survey blocks, each with a 'Show Block' icon and a title followed by the number of questions. The blocks are: 'SIS (2 Questions)', 'Pre-Q (7 Questions)', 'Demographics (Base/Universal) (5 Questions)', 'Demographics (Employment) (5 Questions)', 'SPOKE Vis: Exploration (3 Questions)', 'SPOKE Vis: Tasks (16 Questions)', and 'Post-Q (10 Questions)'. Each block has a set of action buttons: 'Add Below', 'Move', 'Duplicate', and 'Delete'. At the bottom, there is a '+ Add a New Element Here' button.

SPOKE

- ▶ SIS | 2 Questions
- ▶ Pre-Q | 7 Questions
- ▶ Demographics (Base/Universal) | 5 Questions
- ▶ Demographics (Employment) | 5 Questions
- ▶ SPOKE Vis: Exploration | 3 Questions
- ▶ SPOKE Vis: Tasks | 16 Questions
- ▶ Post-Q | 10 Questions

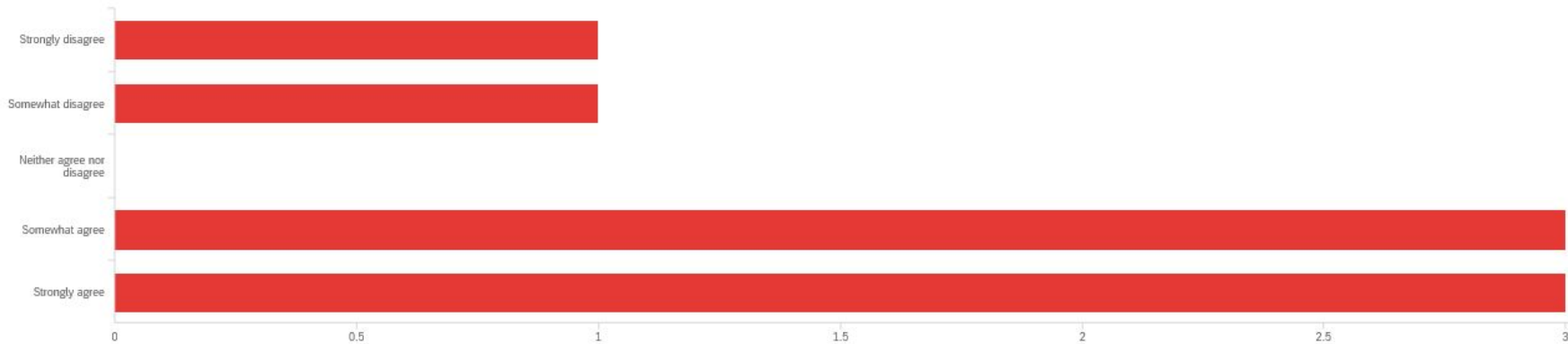
Pilot Study Results: Post-Questionnaire

Q7.4 - Would you say that you found these parts of the documentation we provided for the SPOKE Visualizer helpful?



Pilot Study – Post-Questionnaire

Q7.5 - Would you say that the SPOKE Visualizer was easy to use?



Next steps

- Analyze pilot study data -DONE
- Submit amendment to IRB -DONE
- Write scripts to synchronize Qualtrics with Google Analytics -IN PROGRESS
- Identify patient groups -IN PROGRESS
- Set up Amazon Mechanical Turk -IN PROGRESS
- Log hypotheses in [OSF](#) -IN PROGRESS

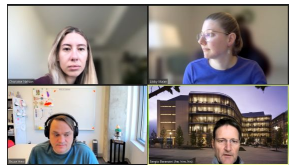
SPOKE-BRIC

Design, Development, and Progress

SPOKE-BRIC Design

How We Collaborate

1. Weekly Meetings
2. Slack
3. Google Sheets
4. Google Docs
5. Figma



Monday, April 4th ▾

Sergio 12:00 PM
I also approved most and commented (just a few) on the Design Review Spreadsheet

Charlotte 5:46 PM
That looks good to me!

👍 1 😊

BRIC MVP Design Review 2022 ☆ 📁 ☁
File Edit View Insert Format Data Tools Extensions Help [Last edit was 5 days ago](#)

BRIC Tooltips ☆ 📁 ☁
File Edit View Insert Format Tools Add-ons Help [Last edit was 5 days ago](#)

MATE Cloudservices

Max # of nodes ?
Total Nodes: 0

Charlotte Nelson Last month
Regarding the orange issues in rows 30 and 34: for the MVP we aren't prioritizing the auto updates here (although I would love to have them if it is possible). If we remove them for the MVP, I would center the max # of nodes section with the show button

Reply

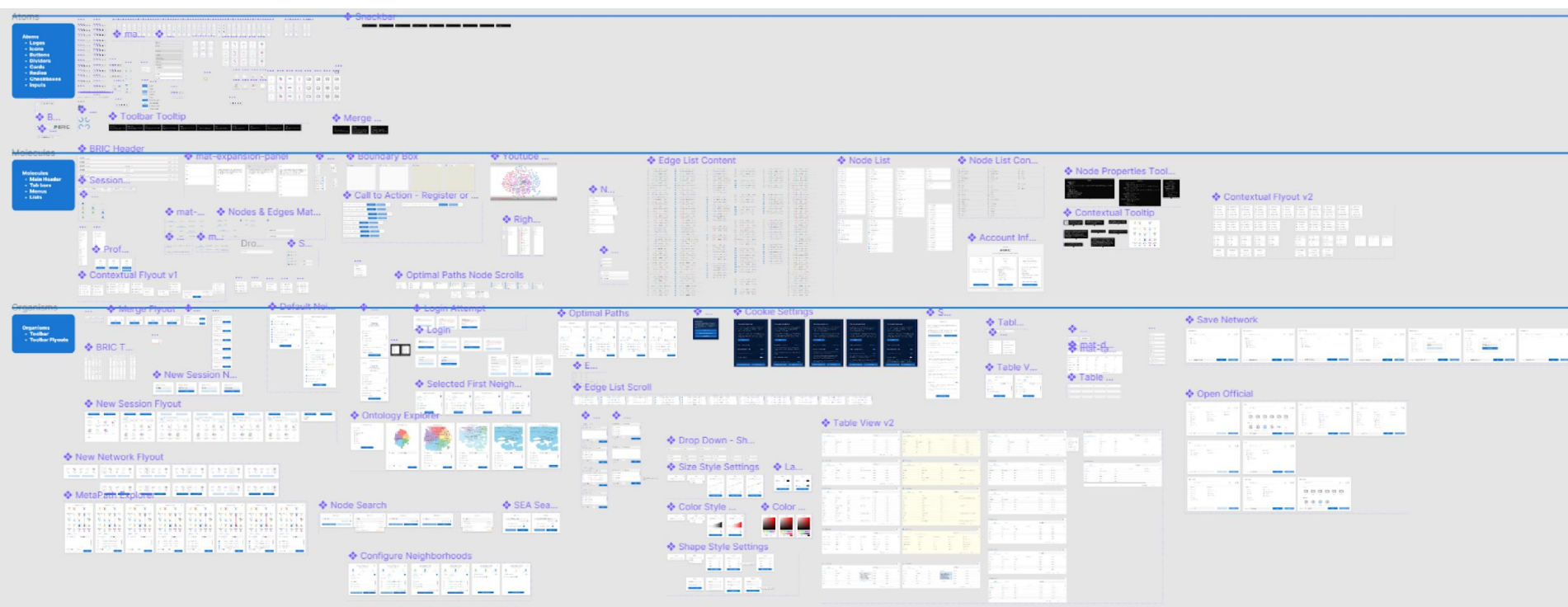


+



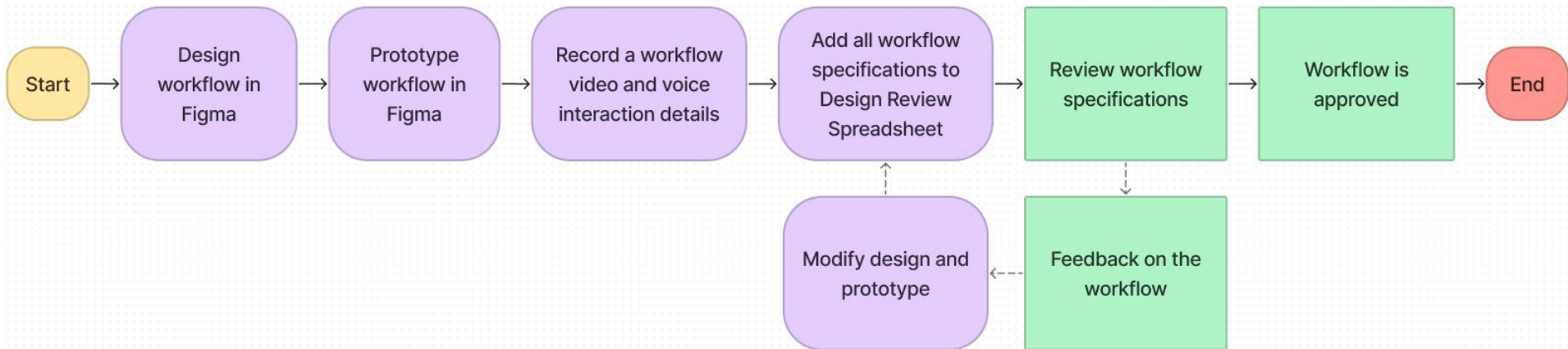
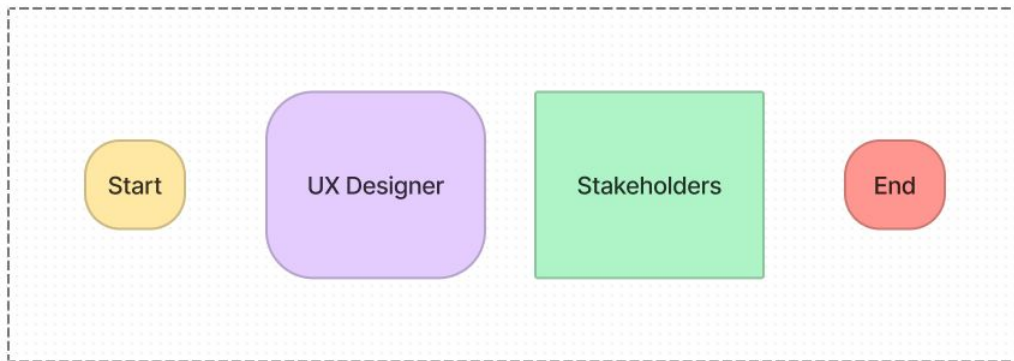
Cloud storage solutions for designing BRIC:

- Promotes transparency during the design process
- Fosters collaboration within cross-functional teams
- Increases efficiency between designers and developers





Minimum Viable Product – Design Review Process





BRIC MVP Design Review 2022							
File Edit View Insert Format Data Tools Extensions Help							
Last edit was seconds ago							
100% 123 Inter 9 B I A 100% 123 Inter 9 B I A 100% 123 Inter 9 B I A							
Do we have a full SEA example? SEB: There are 3 examples of SEA searches in the NE 'Sample queries'.							
	A	B	C	D	E	F	G
1	Flow Demo Video	Interactive Prototype Flow	Design File Flow	Approved (Initials)	Detailed stakeholder comments about what is required for approval:	Additional stakeholder comments	Libby's Comments, To Do's & Progress Updates
2							4/7/1: GitHub updated. Enhancement Issue added. (4/8/1: To do - Implement medium top for landing page and small logo bottom left for app page. Update GitHub. (3/23 post-call: New video. Waiting on feedback. Slacked video and question. MATE logo in the bottom left on app page seems to work but need to confirm. GitHub must be updated eventually. (3/17/1: Waiting on feedback on MATE logo mockups from stakeholders. Video: https://drive.google.com/file/d/1V4n99j0npsShHelyf7zQob5A53npgTF/view?usp=sharing Interactive Prototype (click the MATE icon to see the next available mockup): https://www.figma.com/proto/15nKbyLV03XG9nVg8sChw/BRIC-MVP?mode=id-1166%3A34793487&scale-down&page-id=282%3A3481&starting-point-node-id=282%3A3481&show-proto-sidenav-1)
3	Cookie Notice	Cookie Notice	Cookie Notice	CN/SEB		CN 3/29/1: Like medium up top for the landing page	3/15: Welcome back Snackbar added. View the functioning prototype here: https://www.figma.com/proto/15nKbyLV03XG9nVg8sChw/BRIC-MVP?mode=id-282%3A3481&scale-down&page-id=282%3A3481&starting-point-node-id=282%3A3481&show-proto-sidenav-1
4	Login	Login	Login	CN/SEB	SEB: present a banner "Successful login" or something like this	CN: Works for me CN 3/23: Now that we have the error text below, maybe we don't need to back button? We just need something that prompted them to try again and I think the text is enough.	3/24: Good catch! They can always use the "X" to close out of the login modal should they want to start over. Updated design, prototype, new video, and GitHub flow/issue. Waiting on feedback or approval.
5	Login Attempt	Login Attempt	Login Attempt	CN/SEB	Should there be a back button here? They might have messed up their password	CN 3/23: I'm good with this, but I just want to offer an alternative. When they press the forgot password link, it could just send them to the MATE forgot password page and we could handle the email and reset.	3/24: Libby: v3 video added, GitHub Flow/Issue Updated. Waiting on approval 3/24 BH: Agree with CN, this should send them over to MATE to reset their password.
6	Forgot Password	Forgot Password	Forgot Password	CN/SEB	Maybe is a good idea to confirm their password in a separate box.		4/6: Should be easy to update later. UI is good. (3/23: Waiting on collaboration. Account Information Overlay text added in BRIC Tools. Please update the verbiage. I will update Figma/GitHub when completed.
7	Guest User	Guest User	Guest User	SEB/CN	Does subscribe refer to a paid account? after logging as a guest, should there be a banner or indication that this is a guest account (and that has limitations)? 'clicking outside the youtube modal goes back to previous screen' this is not obvious. Should there be a little banner prompting this?	CN: I think this is really smart. We will just need to update some of the text	https://docs.google.com/document/d/1L6Lmc7BdF1e5hQrOj5jNTp4Wf0eTrB8Uj0eCn0/edit (3/17: Subscribe refers to a paid account. An "X" icon has been added to close out of the YouTube modal and the previous behavior has been removed. Account information overlay designed/prototyped on 3/15. GitHub flow/issue updated 3/17.)
8	Free Accounts	Free Accounts	Free Accounts	CN/SEB			
9	Toolbar Tooltips	Toolbar Tooltips	Toolbar Tooltips	CN/SEB			
10	Settings	Settings	Settings	SEB/CN		CN 3/29: Under the Account tab, I would just have the manage subscription button and have that take them to mate where they can change username/password/subscription all in one place	3/30 This has been completed! Waiting on feedback or approval. View screenshot: https://drive.google.com/file/d/1YOten3cvTgRYCIS1TyOAbUOGwwwdHnZr/view?usp=sharing GitHub Issue/flow is updated. (3/21: Waiting on feedback or approval)
11	New Session	New Session	New Session	CN/SEB			4/5: Completed: New dev video, GitHub flow & Issue. (3/30 Libby post approval TO DO: Must update design file. Creating new video for devs. New GitHub flow/issue (3/23 post call: This was featured on the call today, everyone seemed to approve. Needed after approval. New video and GitHub flow and issue. 3/18: this design proposal is for opening sessions, renaming sessions, and duplicating sessions. Waiting on feedback or approval)
12	Open/Rename/Duplicate Session	Open/Rename/Duplicate Session	Open/Rename/Duplicate Session	SEB/CN		CN: Love it!	
13	New Networks	New Network	New Network	CN/SEB			
14	Node Enrichment	Node Enrichment	Node Enrichment	CN/SEB			
15	Style Settings (Shape, Size, Color)	Style Settings (Shape, Size, Color)	Style Settings (Shape, Size, Color)	CN/SEB			
16	Neighborhood Explorer Node Search, Show Neighborhoods Button & Delete a Network	Neighborhood Explorer Node Search, Show Neighborhoods Button & Delete a Network	Neighborhood Explorer Node Search, Show Neighborhoods Button & Delete a Network	CN/SEB	SEB: can we make the default view be the node-level and make the metagraph level optional by toggle switch? (I'm saying this because I'm mostly using his node level on a daily basis) SEB: Tooltip for 2nd neighbors is not clear. I would just say: "A second neighbor is two hops away from the node of interest". Also, is there a way to make the user aware that the tooltip changes from 1st to 2nd neighbors? (currently there is no way of knowing)		3/18: Absolutely! Network view will be default for all Network tools aside the MetaPath Explorer. This has been noted in the Toggle View flow in GitHub.
17	Neighborhood Explorer Node Search, Selecting 1st & 2nd Neighbors, Clicking Individual Nodes	Neighborhood Explorer Node Search, Selecting 1st & 2nd Neighbors, Clicking Individual Nodes	Neighborhood Explorer Node Search, Selecting 1st & 2nd Neighbors, Clicking Individual Nodes	CN/SEB			3/17: Verbiage has been updated for 2nd neighbors tooltip. Tooltip context has been added next to the question mark icon: "Learn about 1st Neighbors & Learn about 2nd Neighbors". GitHub Issues updated for all Neighborhood Explorer Flows.



Approved Minimum Viable Product

36 interactive prototype workflows

100+ user interface components

348 screen designs



Network Creation Tools

- Ontology Explorer
- MetaPath Explorer
- Neighborhood Explorer
 - Similarity Ensemble Search
 - Node Search
- Optimal Paths
- Node Enrichment



MetaPath
Explorer



Neighborhood
Explorer



Optimal
Paths



Node
Enrichment



Ontology
Explorer

SIMILARITY ENSEMBLE

NODE SEARCH

BRIC

UX/UI Next Steps



SPOKE-BRIC Development

SPOKE-BRIC Development: Crimson Fox Dev Team



sanket1211996

#1

35 commits 3,519 ++ 37,540 --



axdanbol

#2

29 commits 140,808 ++ 41,662 --



bherr2

#3

12 commits 765 ++ 332 --



edlu77

#4

9 commits 2,997 ++ 504 --



adphillips2017

#5

8 commits 2,559 ++ 208 --



Kavyara

#6

7 commits 1,778 ++ 76 --



richajha012

#7

5 commits 824 ++ 6 --



akshaybharadwaj

#8

5 commits 1,091 ++ 27 --



ManognyaKatapally

#9

2 commits 3,470 ++ 11,225 --



sumedhsalvi

#10

2 commits 274 ++ 25 --

+ Bryce Green

SPOKE-BRIC Development Process

— — —

- Atomic Design
- Software Architecture
- Quality Assurance
- Component implementation
- Business Logic
- Minimum Viable Product

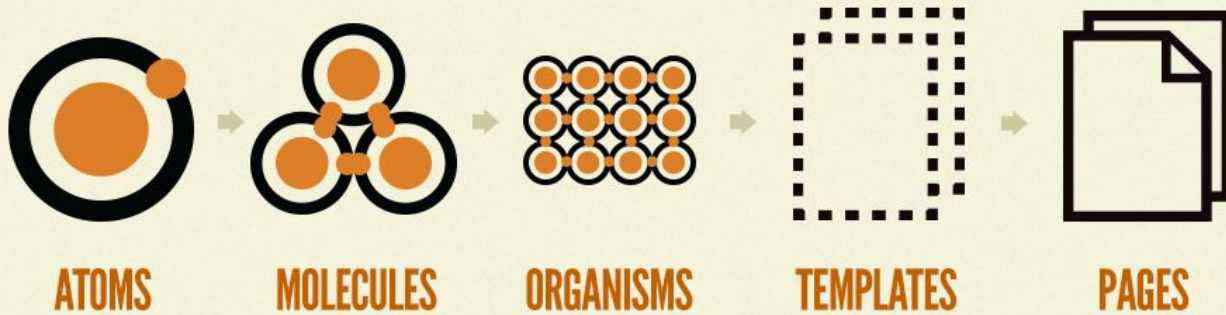
SPOKE-BRIC Development: Atomic Design

— — —

A design system methodology by Brad Frost <<https://bradfrost.com/>> and adopted widely in the industry.

Atomic design is methodology for creating design systems. There are five distinct levels in atomic design: Atoms, Molecules, Organisms, Templates, and Pages.

*The following brown slides are Brad Frost's



SEARCH THE SITE

LABEL

ENTER KEYWORD

INPUT

SEARCH

BUTTON

SEARCH THE SITE

ENTER KEYWORD

SEARCH

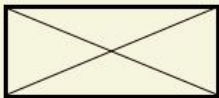


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SEARCH THE SITE

ENTER KEYWORD

SEARCH

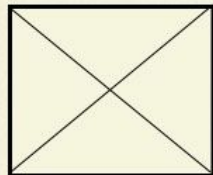
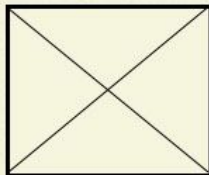
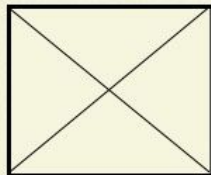
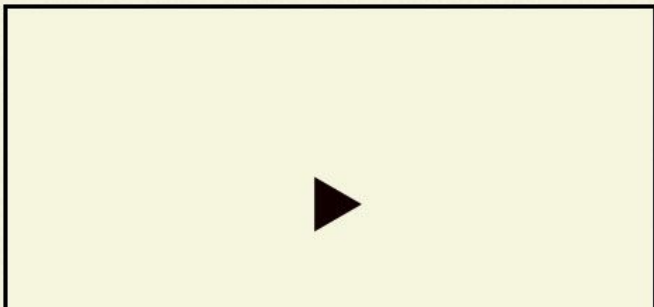
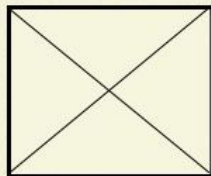
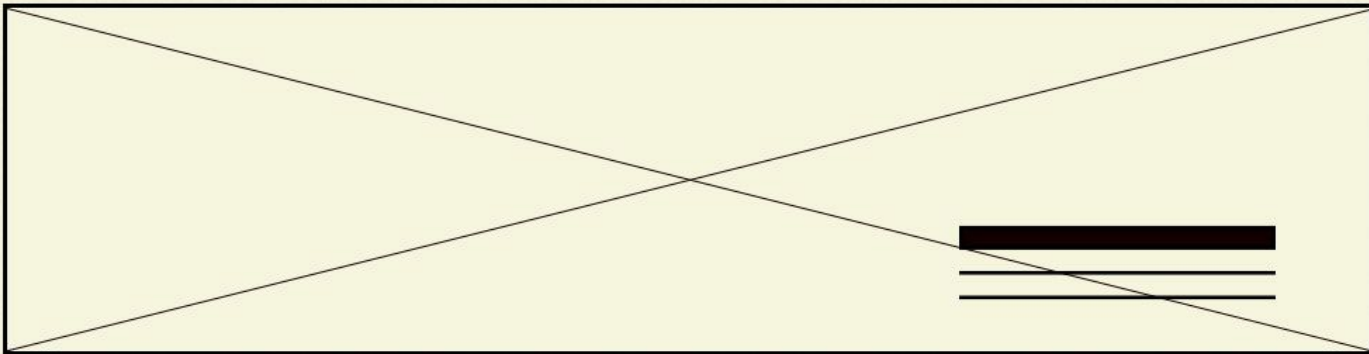


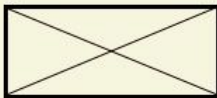
[Home](#) [About](#) [Blog](#) [Contact](#)

SEARCH THE SITE

ENTER KEYWORD

SEARCH





[Home](#) [About](#) [Blog](#) [Contact](#)

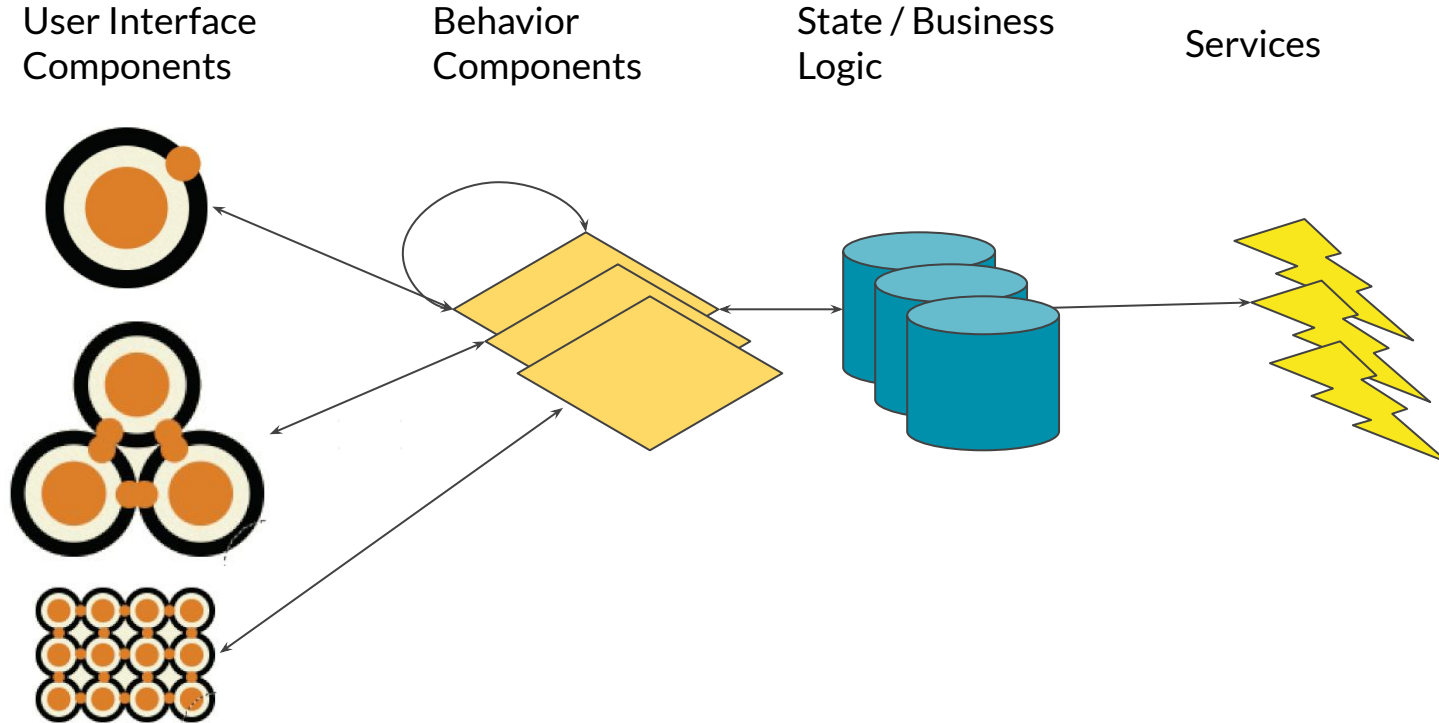
SEARCH THE SITE

ENTER KEYWORD

SEARCH



SPOKE-BRIC Development: Software Architecture



SPOKE-BRIC Development: Software Architecture

— — —

- **UI Components**
 - Pure user interface component building blocks, defined by their @Inputs and @Outputs
 - Atoms, Molecules, and Organisms in Atomic Design
- **Behavior Components**
 - Components which mediate between UI components and the State
 - Might have a little CSS to layout the UI components, but should be mostly unconcerned with UI issues
 - Like organisms in Atomic Design
- **States (ie Business Logic)**
 - State (and transitions) of the application is modeled here
 - Completely separate from UI and can be modeled separately
- **Services**
 - Code to simplify data access or other complex procedures
 - Should be mostly stateless

SPOKE-BRIC Development: Component Implementation

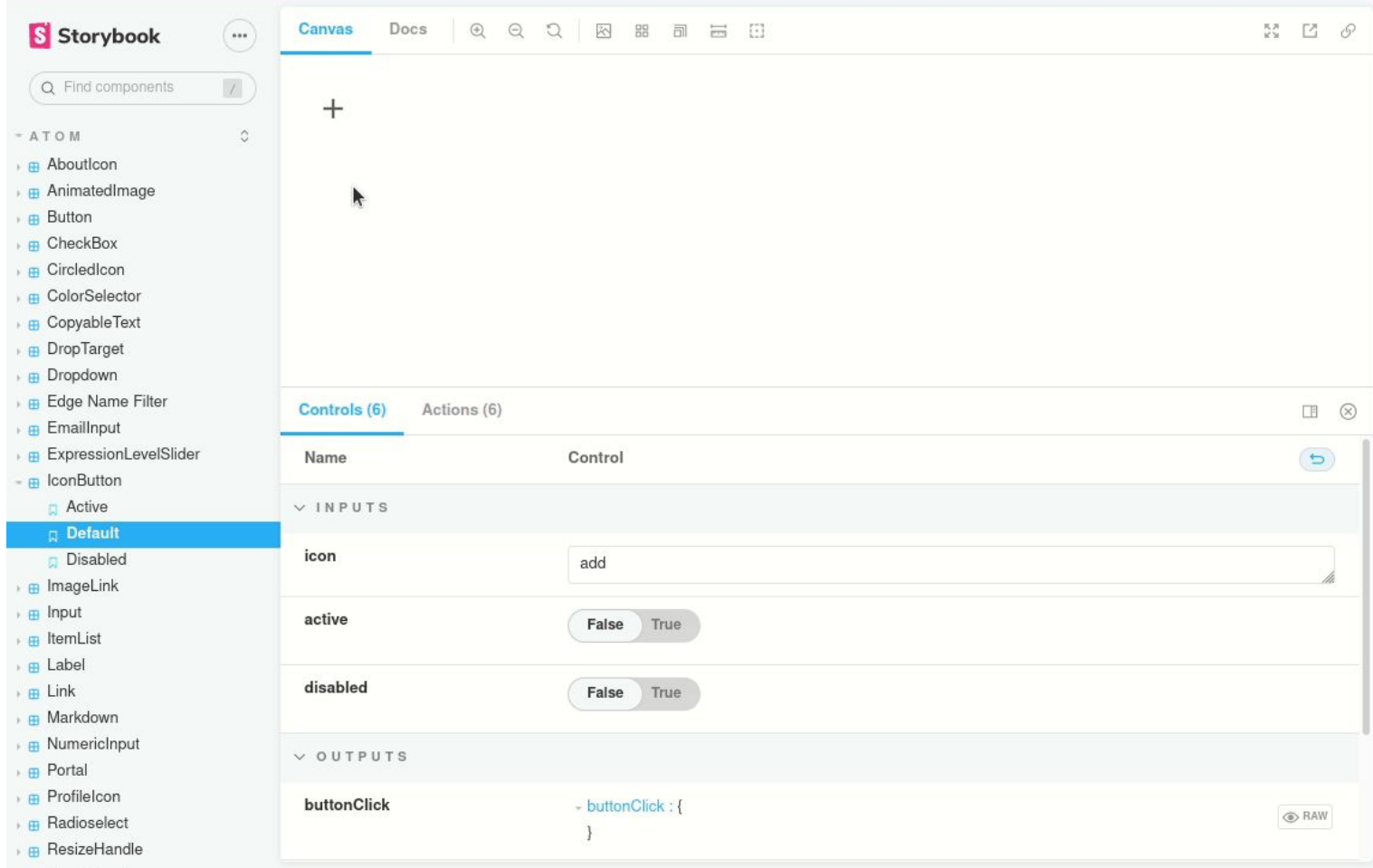
— — —

- Each Flow from the UI Spec is broken down into Atomic design components
- Each component can be worked on and tested in parallel
- Theming from the beginning
- Each component is implemented in the “Angular Way” and best practices we’ve identified

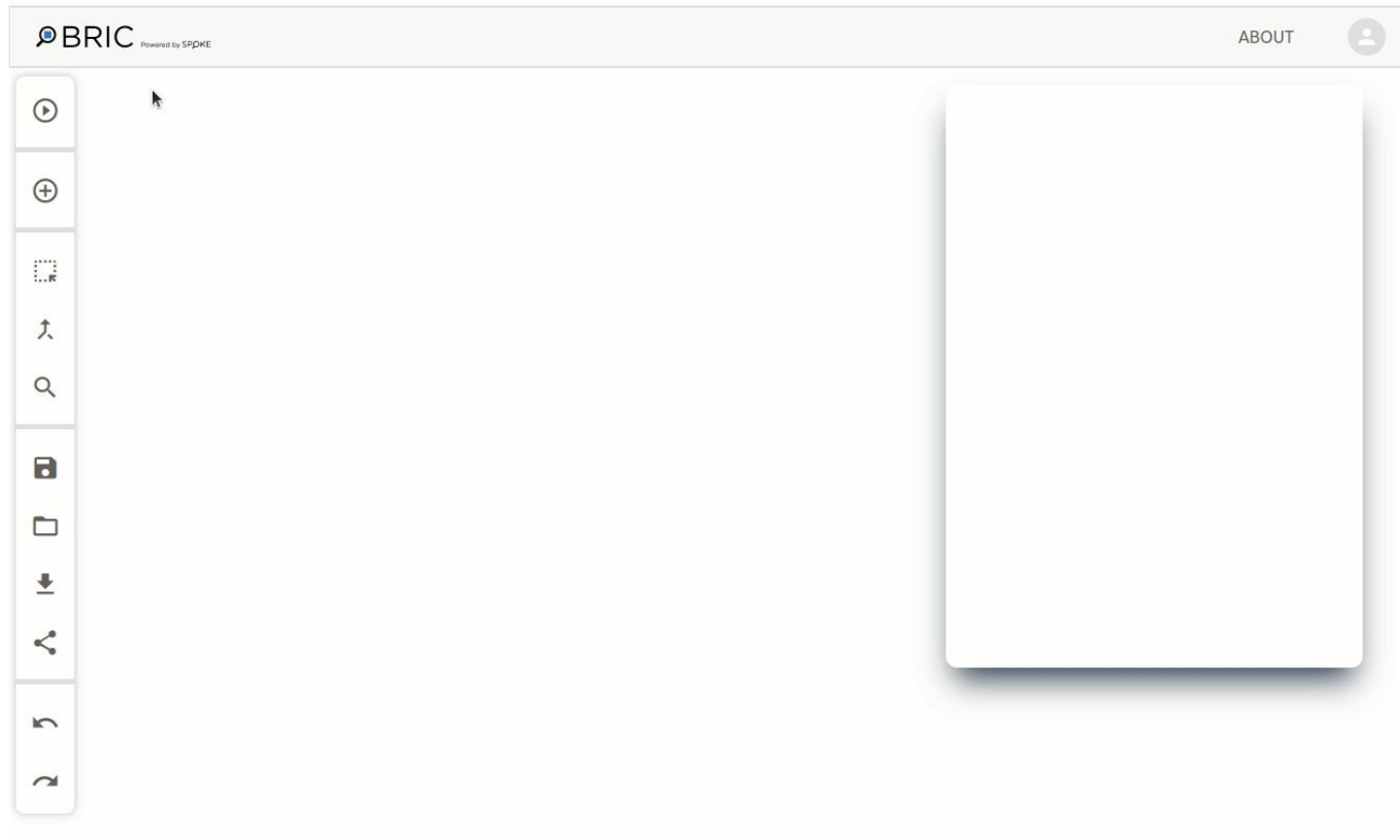
SPOKE-BRIC Development: Business Logic

— — —

- Each Flow reveals the state changes/behavior expected
- This can be translated into a set of States using NGRX
- State is modeled separately from UI to maximize separation of concerns



Storybook: Test, document, and interact with UI Components



Staging site updated as new UI Components come online

SPOKE-BRIC Development: Quality Assurance

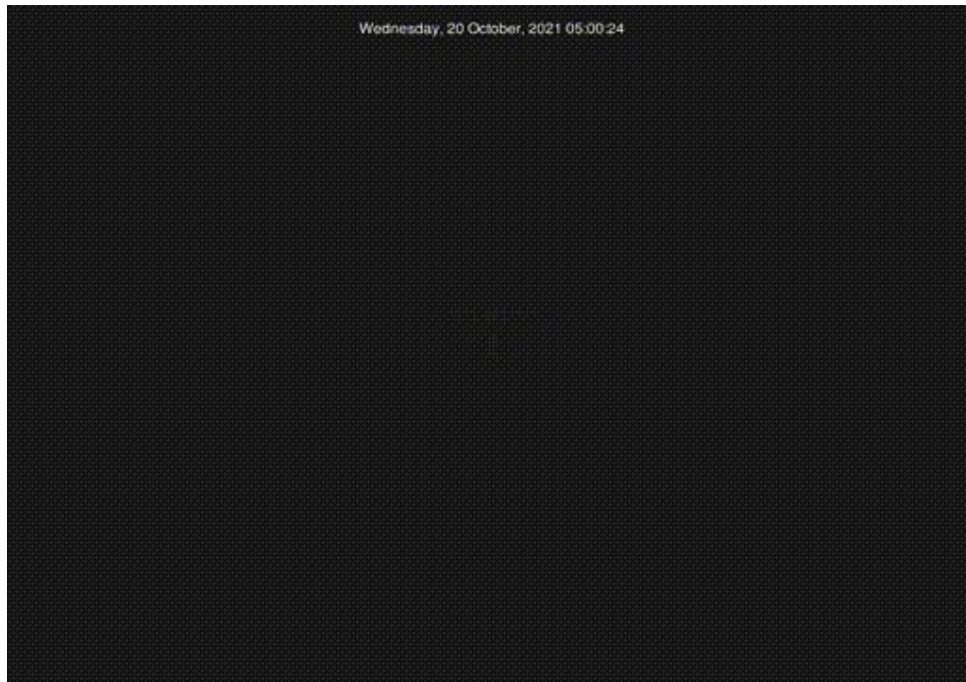
— — —

- 100% Unit Tests
- 100% Documentation coverage
- Storybook for visual feedback and 'interesting states'
- Pull request for each new component/feature
 - CI/CD flow for testing, minimum quality control checks, creates build previews for reviews
 - Code review by senior developers

SPOKE-BRIC Development: MVP

— — —

- 100+ Atomic Design Components identified
- 36 Workflows
- 10 developers all working in parallel
- Still much to learn over the summer, but looking good
- SPOKE-BRIC MVP is due mid-August



SPOKE-BRIC Development: Activity

Total Lines of Code

66,728

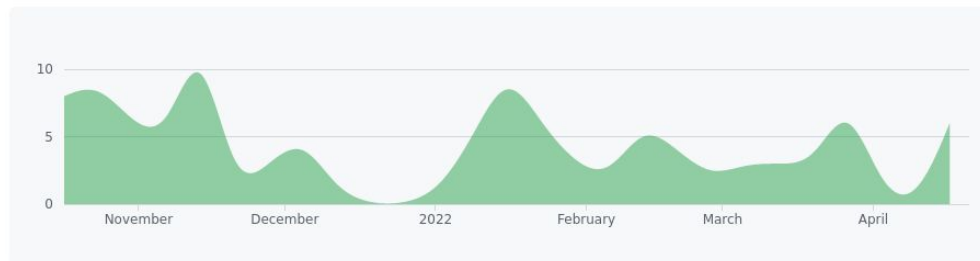
Total Reviewed Commits

123 (average 2.4 commits per active day, 0.7 per all days)

Authors

10 (average 10.2 commits per author)

Reviewed commits to staging per week



	sanket1211996 35 commits 3,519 ++ 37,540 --	#1
	axdanbol 29 commits 140,808 ++ 41,662 --	#2
	bherr72 12 commits 765 ++ 332 --	#3
	edlu77 9 commits 2,997 ++ 504 --	#4
	adphillips2017 8 commits 2,559 ++ 208 --	#5
	Kavyara 7 commits 1,778 ++ 76 --	#6
	richajha012 5 commits 824 ++ 6 --	#7
	akshaybharadwaj 5 commits 1,091 ++ 27 --	#8
	ManognyaKatapally 2 commits 3,470 ++ 11,225 --	#9
	sumedhsalvi 2 commits 274 ++ 25 --	#10

Figma Prototype DEMO